

My Code

First, the beginning I connected our app to an API after having difficulty with Buienradar's API I switched over to OpenWeathers API and this ended up working. I later then got the temperature to display via City names but later changed that to Latitude & Longitude and the city name for additional organization. This allowed us to get data more specifically for the beach and not the city near it.

```

func fetchTemperature() {
    guard let city = selectedCity else {
        print("No city selected")
        return
    }

    guard let url = URL(string:
        "https://api.openweathermap.org/data/2.5/weather?lat=\(city.latitude)&lon=\(city
        .longitude)&units=metric&appid=8c714448edc2ce49f32ce5608cebf590") else {
        print("Invalid URL")
        return
    }

    let task = URLSession.shared.dataTask(with: url) { data, response, error in
        if let error = error {
            print("Error: \(error.localizedDescription)")
            return
        }
        guard let data = data else {
            print("Error: Data is nil")
            return
        }

        print(String(data: data, encoding: .utf8) ?? "")

        do {
            let decoder = JSONDecoder()
            let weatherData = try decoder.decode(WeatherData.self, from: data)
            DispatchQueue.main.async {
                self.temperature = "\(weatherData.main.temp)"
                self.windSpeed = "\(weatherData.wind.speed)"
                self.humidity = "\(weatherData.main.humidity)"
                self.windGust = "\(weatherData.wind.gust ?? 0.0)"
            }
        }
    }
}

```

```

struct ContentView: View {

    @State private var temperature: String = ""
    @State private var windSpeed: String = ""
    @State private var humidity: String = ""
    @State private var windGust: String = ""

    @State private var searchText: String = ""
    @State private var selectedCity: City?

    let cities = [
        City(id: 2747891, name: "Rotterdam", latitude: 51.9225, longitude: 4.47917),
        City(id: 2747373, name: "Den Haag", latitude: 52.0705, longitude: 4.3007),
        City(id: 2759794, name: "Amsterdam", latitude: 52.374, longitude: 4.8897),
        City(id: 2745343, name: "Texel", latitude: 53.0638, longitude: 4.7997),
        City(id: 2744467, name: "Wassenaar", latitude: 52.1483, longitude: 4.4067),
        City(id: 2749877, name: "Middelburg", latitude: 51.5, longitude: 3.61389)
    ]

    var filteredCities: [City] {
        if searchText.isEmpty {
            return cities
        } else {
            return cities.filter { $0.name.lowercased().contains(searchText.lowercased()) }
        }
    }
}

```

Later I created a Livestream view to display a live stream from youtube, this is to allow our users to

be able to see a beach live and see the waves themselves. I tried to fiddle around with some of the design of the youtube embed but it always seems to look wrong with the rest of our app.

```
//
import SwiftUI
import WebKit

struct WebView: UIViewRepresentable {
    let url: URL

    func makeUIView(context: Context) -> WKWebView {
        let webView = WKWebView()
        webView.load(URLRequest(url: url))
        return webView
    }

    func updateUIView(_ webView: WKWebView, context: Context) {}
}

struct YoutubePlayerView: View {
    var body: some View {
        WebView(url: URL(string: "https://www.youtube.com/embed/EHK0x5QJEj4?controls=1&showinfo=0")!)
            .frame(width: 560, height: 315)
    }
}
```

I also added an arrow that could change direction based on the wind's direction. This really helped

with our design and made the wind radar more usable.

```
        VStack {
            HStack{
                Text("Wind")
                    .font(.title3)
                    .fontWeight(.medium)
            }
            Text("Direction")
                .font(.title3)
                .fontWeight(.medium)
            Image(systemName: "location.north.line")
                .foregroundColor(.blue)
                .rotationEffect(.degrees(Double(windDegree) ?? 0))
        }
        .padding()
        .frame(width: 150, height: 100)
        .background(
            RoundedRectangle(cornerRadius: 20)
                .fill(Color.white)
                .shadow(color: .gray, radius: 5, x: 0, y: 2)
        )
    }

    RadarView()

    Spacer()

}

.onAppear {
    self.fetchTemperature()
}
```